

University of Porto

CC3046 - Introduction to Intelligent Robotics

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Project Proposal

Group T3_G9

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**Vision-Based Autonomous Pursuit-Evasion
in Urban Environments with Recurrent Policies**

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Summary of the goals of the work

While classical motion planning algorithms perform well in static environments with known maps, they struggle in dynamic and partially observable scenarios. Moreover, real-world robotic systems cannot rely on shared global information and must act based on their own perception. Therefore, this project explores reinforcement learning techniques, specifically Proximal Policy Optimization (PPO), within the “Sense-Think-Act” paradigm in an urban simulation environment. The main objective is to train an autonomous **evader vehicle** capable of avoiding a pursuing agent in a “cops and robbers” scenario in the Webots simulator. The pursuer will follow a predefined or reactive strategy, acting as a dynamic opponent rather than a learning agent. Using only onboard sensors (camera and LiDAR), the evader must learn to navigate safely, break line-of-sight, and handle temporary loss of visual information. The main goals are:

- Development of an evasion system operating under partial observability. Although the evader relies solely on local perception, the system introduces information asymmetry where the pursuer receives the evader’s global location (GPS) at a defined frequency X .
- Design of a reward function that encourages survival, distance from the pursuer, and line-of-sight breaking.
- Implementation of PPO with LSTM to provide short-term memory for handling visual occlusions.
- Development of safe and effective evasive behaviours combining navigation and obstacle avoidance.
- Evaluation of performance using metrics such as survival time, success rate, and collision rate.

As an optional extension, we may explore a more complex setup where both the pursuer and the evader are trained using reinforcement learning.

Proposed approach to solve the problem

Description of the setup

This project will be fully developed on the robotic simulator Webots. The simulator provides a virtual environment where autonomous vehicle agents can interact with each other and the urban infrastructure. It leverages a physics engine to simulate vehicle dynamics, collisions, and sensor behaviour. A 3D visualisation tool allows real-time observation of the agents. The environment used is the predefined “City Traffic” world. The robots are equipped with sensors and actuators to perceive and act upon the world. These include steering and throttle control, cameras for detecting the pursuer, LiDAR for obstacle perception, and contact sensors for detecting collisions. Additionally, the environment simulates a tracking system where the pursuer vehicle receives the evader’s GPS coordinates at a predefined update rate (frequency X), simulating support from a radar or radio central. The environment includes an urban layout with roads, intersections, and buildings. Static obstacles include buildings and sidewalks, while the pursuer vehicle acts as a dynamic obstacle. The system operates under partial observability, requiring the evader to rely only on its own perception.

Algorithms

The proposed techniques are Proximal Policy Optimization (PPO), Long Short-Term Memory (LSTM), reward shaping, and a predefined pursuer strategy. The focus is on training only the evader, keeping the problem manageable.

Proximal Policy Optimization

PPO will be used to train the evader agent. It provides stable learning and is suitable for continuous control tasks such as vehicle navigation.

Long Short-Term Memory (LSTM)

LSTM layers will be added to the policy to handle partial observability. This allows the evader to use recent past information when the pursuer is temporarily not visible.

Reward Shaping

The evader will be rewarded for:

- Increasing distance from the pursuer
- Breaking line-of-sight
- Surviving longer

It will be penalized for:

- Being captured
- Collisions with the environment
- Staying too close to the pursuer

The episode ends when the evader is captured or when a time limit is reached.

Pursuer Strategy

The pursuer will follow a predefined behaviour that leverages asymmetric information. It will receive the evader's exact location (via GPS) at a frequency X . Between GPS updates, the pursuer will navigate towards the last known position. When the evader enters its field of view, the pursuer will transition to a purely reactive direct pursuit behaviour, while using simple obstacle avoidance. This provides a consistent opponent during training.

Proposed experiments to evaluate the quality of the approach

The trained evader policy will be evaluated over multiple runs with randomized initial conditions.

Experiment 1 – Evasion Performance

- **Setup:**
 - The evader must survive while being pursued.
 - The pursuer follows a predefined strategy.
 - 100 runs with random initial positions.
- **Metrics:**
 - Average Survival Time
 - Escape Success Rate
 - Collision Rate

Experiment 2 – Line-of-Sight Breaking

- **Setup:**
 - Evaluate how the evader uses the environment to break visual contact.
 - 100 randomized runs.
- **Metrics:**
 - Stealth Efficiency
 - Number of successful line-of-sight breaks
 - Re-detection frequency

Experiment 3 – Effect of Memory (Optional)

- Compare PPO with LSTM vs PPO without LSTM.
- Evaluate impact on survival and stealth.

Experiment 4 – Impact of GPS Update Frequency (Optional)

- **Setup:**
 - Evaluate how the evader’s global location update rate affects pursuit dynamics and evasion performance.
 - Test the trained evader against a pursuer configured with different GPS update frequencies X (e.g., updates every 2, 5, and 10 seconds).
- **Metrics:**

- Correlation between GPS update frequency and Average Survival Time.
- Variation in Escape Success Rate based on the pursuer's informational advantage.

Milestones Outline

Weeks Leading up to the Checkpoint (Weeks 1-7)

Week 1 - 2 (27/03 - 10/04): Initialization and Simulation Setup

1. Set up the Webots simulation environment and configure the “City Traffic” world.
2. Conduct preliminary sensor tests (Camera, LiDAR, and Bumper).
3. Define action space and respawn logic.

Week 3 - 5 (10/04 - 01/05): PPO Implementation and Initial Training

4. Implement PPO for the evader and design the reward function.
5. Implement the pursuer baseline behaviour.
6. Run initial training in simplified scenarios.
7. Begin drafting the report methodology.

Week 6 - 7 (01/05 - 14/05): LSTM Integration and Checkpoint Preparation

8. Integrate LSTM into the policy.
9. Refine reward shaping.
10. Prepare checkpoint demonstration.

Objectives for Intermediate Delivery (14/05)

- Working PPO implementation for the evader.
- Basic pursuer behaviour.
- Demonstration of initial evasion behaviour.
- Draft report (introduction and methodology).

Weeks Leading up to the Final Delivery

Week 8 - 10: Experiments and Analysis

11. Run experiments and collect performance data, including variations in GPS frequency.
12. Analyse behaviour, results, and adaptation to different pursuit difficulty levels.

Week 11 - 12: Finalization

13. Finalize report and conclusions.
14. Code cleanup and testing.
15. Prepare presentation.

Objectives for Final Delivery

- Final project submission (code and report).
- Completed experiments and analysis.
- Oral presentation and demonstration.
- Optional exploration of a two-agent reinforcement learning setup.